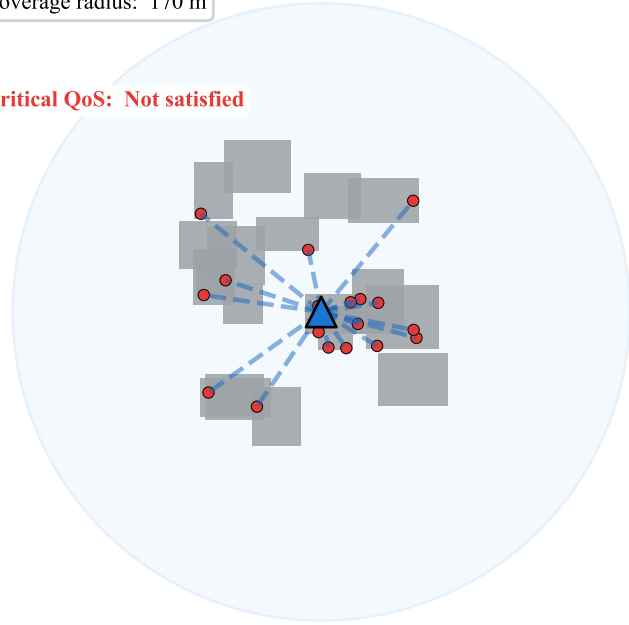


# Altitude → Coverage Radius → Communication Quality

2D-AUTO

Altitude: 100 m  
Coverage radius: 170 m

Critical QoS: Not satisfied

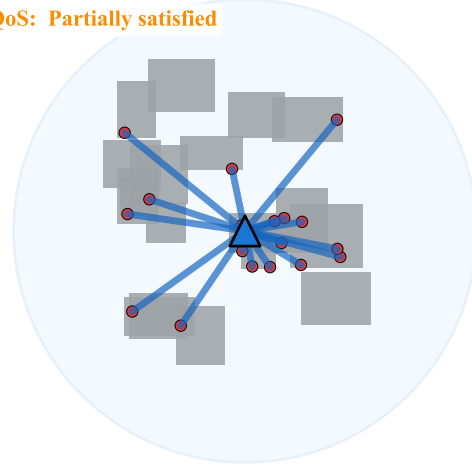


Large footprint  
Low link quality

3D-GNN

Altitude: 75 m  
Coverage radius: 127 m

Critical QoS: Partially satisfied

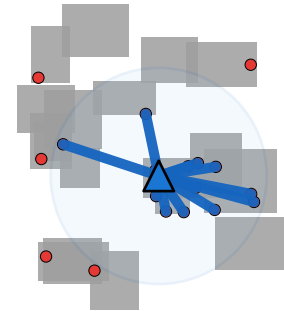


Moderate footprint  
Limited adaptation

ATOM-3D-VoI (Ours)

Altitude: 35 m  
Coverage radius: 59 m

Critical QoS: Satisfied



Focused footprint  
High QoS for critical nodes

● High priority    ● Medium priority    ● Low priority    ▲ UAV    — Critical link (thicker = higher quality)    Coverage footprint